

2024



ATAL Helps Building Developers Save Up to 15% on Annual Energy Costs with Analytics Platform on AWS

ATAL Engineering Group transformed its data analytics platform using Amazon EMR and Amazon Athena, helping customers reduce energy consumption through new insights.

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200%
increase in analytics performance

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500%
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80% Overview

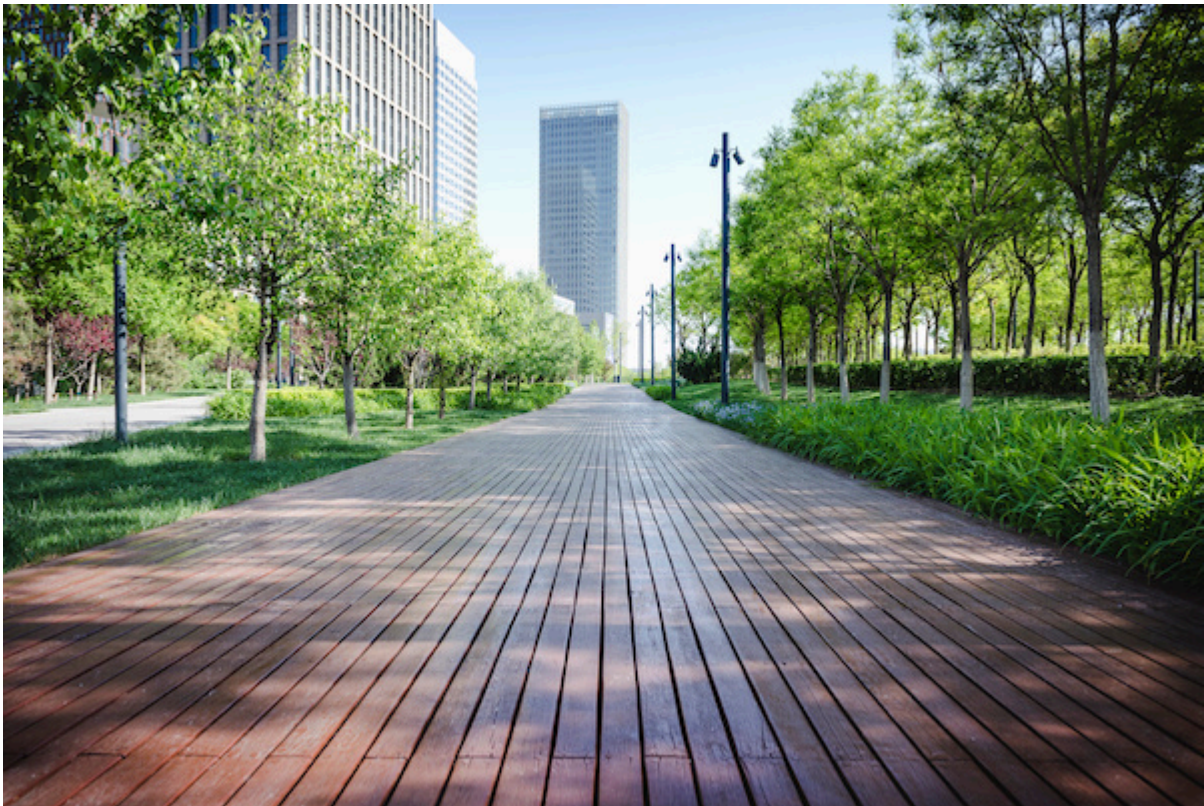
decrease in downtime

7-15%

annual energy cost savings

ATAL's Information, Communications & Building Technologies (ICBT), a business segment of the Hong Kong-based [ATAL Engineering Group](#) (ATAL), provides data-driven solutions for sustainable building management. To reduce management time and allow its developers to focus on building sustainable solutions, ICBT migrated its on-premises analytics platform to Amazon Web Services (AWS).

The platform uses [Amazon EMR](#) for big data processing, [Amazon Athena](#) for data querying, and [Amazon QuickSight](#) for business intelligence. ICBT also deployed [Amazon SageMaker Canvas](#) for rapid machine learning proofs of concept. As a result, the division boosted analytics performance by 200 percent and has helped customers save 7–15 percent per annum on its energy consumption.



Opportunity | Helping Developers Manage Buildings More Sustainably

[ATAL Engineering Group](#) (ATAL) is one of the leading electrical and mechanical engineering service providers in Hong Kong. Established in 1977, the group delivers multi-disciplinary and comprehensive engineering and technology services to a wide spectrum of public- and private-sector customers.

One of the group's key divisions is ATAL's Information, Communications & Building Technologies (ICBT). ICBT offers design, installation, and servicing for intelligent systems and green building solutions. It specializes in energy optimization, fault detection and design, and heating and air conditioning systems.

As developers seek to make buildings more environmentally friendly, ICBT's technical team has been at the forefront of creating sustainability solutions. One of ICBT's first projects was its Dynamic Energy platform, which analyzes data from buildings using machine learning (ML) models. Administrative teams use the platform to manage heating, lighting, ventilation, and maintenance schedules more efficiently.



However, the ICBT technical team overseeing the on-premises infrastructure behind the platform found themselves allocating more time to maintenance tasks rather than enhancing its capabilities. "We want to improve the value of ATAL by prioritizing software development over fundamental infrastructure tasks," says Pan Lee, ICBT technical manager at ATAL.

Solution | Seamlessly Integrating AWS to Enhance Analytics and ML Modeling

ICBT decided to develop an easy-to-manage cloud solution for advanced analytics and ML modeling on Amazon Web Services (AWS). "We chose AWS because of the maturity of its solutions and the availability of AWS developers in Hong Kong," explains Lee.

To store data for analysis, ATAL implemented a Delta Lake open-source data lakehouse. Working closely with AWS, team members integrated the data lakehouse with [Amazon Simple Storage Service](#) (Amazon S3), an object store designed to retrieve any amount of data from anywhere; [Amazon EMR](#) for big data processing; and [Amazon Athena](#), a serverless, interactive analytics service to analyze petabyte-scale data and uncover new insights.

ATAL also uses [Amazon QuickSight](#) as a business intelligence tool for modern interactive dashboards and reporting. Comments Lee, "The AWS team trained us on Amazon QuickSight, illustrating their ongoing technical support throughout the project and further reinforcing our satisfaction with AWS."

Upon migrating its Dynamic Energy Optimization platform to AWS, ATAL saw a 200 percent increase in the platform's performance. Unlike the on-premises solution, which took 2–3 days to transform and load a year's worth of data for pre-processing analysis, the AWS platform completes the same task, including processing, in just 1–2 hours. "If we need to analyze data over several years, we can maintain the same level of performance by scaling up the AWS platform," says Alan Ng, assistant technical manager at ICBT.

In addition, ATAL has reduced infrastructure management time by 50 percent, cut downtime by 80 percent, and minimized data loss. "With AWS, we can prioritize ML development and improve our solutions," says Lee. "This will ensure customers can operate their buildings in more efficient, energy-saving ways."

For ML modeling, ATAL implemented [Amazon SageMaker Canvas](#). While the ICBT business segment was previously using Jupyter Notebook hosted on premises for ML algorithm development, it now has a cloud-based solution to launch



ML proofs of concept (POCs) in less than one day compared to over a week previously. “With Amazon SageMaker Canvas, we gained the agility to launch POCs quickly without having to write a single line of code,” says Lee.

Outcome | Achieving 7–15% Annual Energy Cost Savings for Customers

By using insights from the analytics platform on AWS, the ICBT R&D team is helping customers operate their buildings more sustainably. ATAL can now integrate data from entrance turnstiles and other standalone systems with its main building management systems.

Once processed and analyzed by ML models, this data can provide customers with precise information on reducing emissions related to their buildings. This includes fine-tuning heating and ventilation systems during the day as occupancy rates rise and fall. “With our analytics and ML capabilities on AWS, we’re helping customers save 7–15 percent on their energy costs per building,” says Lee.

Lee continues, “Working closely with AWS, we’ve successfully harnessed technology to contribute to our environmental, social, and governance goals. By transforming our analytics platform, we’re not only achieving significant energy cost savings but also advancing sustainability in the construction and real estate industry.”

These results are generating interest from potential customers that want to improve the environmental record of their commercial properties. “We’re currently talking to several developers who have seen what our Dynamic Energy Optimisation platform can achieve,” Lee comments.

Looking ahead, ATAL plans to switch from hosted Jupyter Notebooks to [Amazon SageMaker](#) for developing ML models. This will help the team automate processes like ML pipeline management, model training, and model storage. “By adopting AWS managed services for our analytics platform, we’ve increasingly shifted our focus to development. We aim to replicate the same benefits by using Amazon SageMaker for upcoming ML initiatives,” concludes Lee.

About ATAL Engineering Group (ATAL)

Headquartered in Hong Kong, ATAL Engineering Group (ATAL) is a leading electrical and mechanical (E&M) engineering service provider with operations in Macau, Mainland China, the United States, and the United Kingdom. Serving a wide spectrum of customers from public and private sectors, the group provides multi-disciplinary and comprehensive E&M engineering and technology services in four major segments, including Building Services; Environmental Engineering; Information, Communications and Building Technologies (ICBT); and Lifts and Escalators.

AWS Services Used

Amazon EMR



Amazon EMR is the industry-leading cloud big data solution for petabyte-scale data processing, interactive analytics, and machine learning using open-source frameworks such as [Apache Spark](#), [Apache Hive](#), and [Presto](#).
[Learn more »](#)

Amazon Athena

Amazon Athena is a serverless, interactive analytics service built on open-source frameworks, supporting open-table and file formats. Athena provides a simplified, flexible way to analyze petabytes of data where it lives.

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Amazon QuickSight

Amazon QuickSight powers data-driven organizations with unified business intelligence (BI) at hyperscale.

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Amazon SageMaker Canvas

SageMaker Canvas provides access to ready-to-use models including foundation models from [Amazon Bedrock](#) or [Amazon SageMaker JumpStart](#) or you can build your own custom ML model.

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